

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The client is DriverPass
- DriverPass aims to take advantage of a void in the market related to training students for their local department of motor vehicles (DMV) driving tests.
- By providing students preparing for driving tests with adequate training to help them successfully pass their tests.
- Provide a web-based system that allows students to access training materials, schedule driving lessons, and track their progress.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass has identified a gap in driver training prior to driving tests at the DMV leading to many drivers failing their tests.
- DriverPass is looking to establish a company that fills this gap by providing students with adequate training to increase odds of passing.

- DriverPass wants a system that is accessible online from any computer or mobile device.
- DriverPass wants a system with defined roles for permissions and access:
 - Owner access / Lead IT access
 - Secretary access
 - Customer access
- This system requires several components to operate, including:
 - Online classes
 - Practice tests
 - On-the-road training

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The system shall be able to create and manage client profiles
- The system shall be able to create, modify, and cancel reservations
- The system shall assign a car and driver to each reservation
- The system shall support user roles with defined permissions:
 - Owner and Lead IT staff shall have full access to all accounts and system functions.
 - Shall be able to reset account passwords as needed.
 - Shall be able to disable training packages that are no longer active or relevant.
 - Secretary shall be able to access, create, modify or delete reservations for all accounts
 - Customer shall be able to access, create, modify or delete reservations for their own account only
- The system shall offer training packages and allow selecting a package based on customer needs.
- The system shall track the creation, modification or cancellation of reservations, including the user responsible.
- The system shall allow users to select specific dates and times for driving lessons
- The system shall collect and store a pickup location for each driving lesson and automatically assign the same location as the drop-off location.
- The system shall track cars, drivers and lesson times to support efficient scheduling.
- The system shall have several pages including:
 - A home page matching the picture description provided
 - An online test progress page
 - A driver notes page, including:
 - Lesson time
 - Start hour
 - End hour
 - Driver comments
 - Assigned car and driver information
 - A personal information page that stores student information and allows updates by student or authorized staff.
 - An input form for adding new students
 - A contact page for communication with the company

- The system shall be accessible on desktop and mobile devices, and utilize cloud infrastructure for storage, backups, and security.
- The system shall support updates and notifications related to DMV rule changes to ensure training materials remain up to date.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The DriverPass system shall operate as a web-based system, accessible through standard web browsers on desktop and mobile devices. No standalone desktop or mobile application is required. Users must be able to access the system from any location with an internet connection, ensuring flexibility for customers, secretaries, administrators, and IT staff.
- The system shall load all pages within 2 seconds under normal operating conditions and no longer than 3 seconds during peak usage. The system shall respond to user actions, such as logging in, scheduling appointments, viewing lesson information, and accessing online tests, within 1 second and no longer than 2 seconds during periods of high usage.
- The system shall be updated regularly as needed, when changes occur to DMV rules, policies, or practice test content. Updates to system functionality, security, and content should be deployed centrally so that all users automatically access the most current version without requiring manual updates or installations.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The DriverPass system shall be platform-independent and accessible through a standard web browser on all desktops and mobile devices. No client-side software installation should be required.
- The back-end system shall use a **SQL-based database** to store customer data, employee data, appointments, packages, and audit logs. The system shall be hosted in a **cloud-based environment** to reduce the client's responsibility for system backups, security, and infrastructure maintenance.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Each user shall be uniquely identified by a system-generated user ID and assigned a role (e.g., Administrator, IT Officer, Secretary, Customer). Role-based access control shall ensure users can only perform actions appropriate to their role.
- User input shall not be case-sensitive where appropriate (e.g., usernames and names).
- The system shall maintain an audit trail for administrative staff that records who created, modified, or canceled appointments, as well as any changes made to system records.
- Administrators must be notified of system activity and be able to generate reports showing all activity, including reservation changes and account updates, to ensure accountability and traceability.
- Administrators shall also be informed when system management actions are required, such as significant record changes or maintenance-related events.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system shall allow administrators to **add, remove, or modify users and accounts** without requiring changes to the system's source code. Secretaries shall be able to add, remove, or modify user appointments. Customers shall be able to update their own account information as permitted by their role and add, modify or cancel their appointments.
- The system shall be flexible enough to allow administrators to disable training packages that are no longer relevant without modifying code.
- As a web-based system, the application shall adapt to browser and platform updates with minimal impact.
- IT administrators must have full access to maintain the system, manage accounts, apply updates, and monitor system functionality.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users shall be required to authenticate using a **username and password** to access the system. All sensitive user data, including personal and payment information, must be stored securely using encryption.
- All data sent between the user's browser (client) and the DriverPass server will be transmitted over HTTPS, with system communication implemented through RESTful web services to ensure secure and structured data exchange.
- The system shall limit login attempts to prevent brute-force attacks by locking an account after a predefined number of failed login attempts.

- If a user forgets their password, the system must support a secure password reset process, either through administrative intervention or an automated reset mechanism after identity verification.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall validate user credentials when logging in.
- The system shall restrict access to system features based on assigned user roles.
- The system shall allow users to reset their password.
- The system shall create, view, update, and deactivate customer profiles.
- The system shall allow customers to register for an account and manage their own profile information.
- The system shall allow customers to create, modify, and cancel their own driving lesson reservations.
- The system shall allow administrative staff (e.g., secretary, admin) to create, modify, and cancel reservations on behalf of customers.
- The system shall allow users to select specific dates and times for driving lessons.
- The system shall assign an available car and driver to each reservation based on the selected date and time.
- The system shall prevent scheduling conflicts by blocking double-booking of cars, drivers, or time slots.
- The system shall collect and store a pickup location for each driving lesson and set the drop-off location to match the pickup location by default.
- The system shall offer training packages for purchase/selection and associate the selected package with the customer account.
- The system shall record reservation activity (create/modify/cancel) including the user responsible and the time of the action.
- The system shall allow administrative users to enable or disable training packages so that customers cannot register for inactive packages.
- The system shall generate activity reports for reservation activity.
- The system shall allow customers to access online practice tests.
- The system shall allow administrative users to export reports and system data for offline use.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The DriverPass system shall provide a web-based, role-based interface accessible through standard browsers on desktop and mobile devices. Customers shall be able to manage their accounts, schedule or modify driving lessons, and view test progress through the interface. Administrative users shall be able to manage reservations, users, vehicles, and reports using a clear and intuitive layout.

- The users of the system include, Owner/Admin, IT admin, secretary, customer
- Owner/Admin
 - The interface shall allow the Owner/Admin to view and manage customer accounts, employee accounts, reservations, vehicles/drivers, training packages, and reporting.
 - The interface shall allow the Owner/Admin to view audit/activity reports showing who created, modified, or canceled reservations.
- IT Admin
 - The interface shall allow the IT Admin to manage user accounts, reset passwords, and disable user access when needed.
 - The interface shall allow the IT Admin to access system maintenance options (as appropriate) and view system activity logs.
- Secretary
 - The interface shall allow the Secretary to create, modify, and cancel reservations for customers and view schedules for cars/drivers.
- Customer
 - The interface shall allow customers to register, log in, manage their profile, select a package, and create/modify/cancel their own reservations.
 - The interface shall display online test progress information (e.g., test name, time taken, score, and status such as not taken/in progress/failed/passed) and lesson/driver notes where applicable.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Due to the limited scope of the project and agreed-upon design compromises, it is assumed that the project operates under practical time and budget limitations, even though these constraints were not explicitly stated in the interview. Certain aspects of the system were not fully addressed in the design, including the process for integrating DMV updates, whether those updates occur in real time or through periodic manual updates, and how system notifications are delivered to administrators when changes or actions require attention. These omissions may impact how quickly regulatory changes are reflected in the system and how effectively administrators respond to system events.
- The design assumes that all users have consistent internet access and access to a modern device capable of running a current web browser. It also assumes that customers and staff possess basic computer literacy, allowing them to navigate a web-based scheduling and testing system without extensive training. Additionally, the design assumes that users enter accurate and complete information, such as pickup locations, availability, and contact details, and does not include advanced validation beyond basic input checks. From a technical standpoint, the design assumes that secure authentication mechanisms and supporting security infrastructure are available and managed by the hosting provider.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The system design is limited by defined project scope and agreed-upon design boundaries, which restrict the scope to essential scheduling, account management, and testing functionality. Advanced features such as real-time DMV integration, automated payment reconciliation, and extensive customization of training packages are outside the scope of the current design. The system also does not support offline functionality, as all data access and modification require an active internet connection.
- Additionally, system scalability and performance under heavy usage have not been fully addressed and may require future optimization as the user base grows. Any expansion of features or improvements in performance would require additional development time, technical resources, and potential changes to the system architecture.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

